GARV ARORA

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Autonomous Systems & Applied AI Engineer | ROS2 · Computer Vision · Machine Learning · Embedded Systems · Python/C++

EDUCATION

Vellore Institute of Technology Aug 2023 - Jul 2027
B.Tech, Computer Science & Engineering (Specialization: IoT) | CGPA: 8.36/10.0
GEMS International School, Gurugram, India Nov 2016 - Apr 2023
CBSE — Class 12: 76.4% | Class 10: 89.3%

EXPERIENCE

Research & Development Lead — SEDS Projects VIT (SEDS India) Jan 2026 - Present
- Promoted to R&D Lead overseeing 2 competition teams (50+ engineers) — Team Ardra (ISDC) and Team Vyadh (IRC/IRDC); resolved 10+ critical pre-competition issues.

Autonomous Systems Developer — Team Vyadh, SEDS Projects VIT Apr 2024 - Jan 2026
- Architected a full ROS2 navigation stack (Nav2, RTAB-Map Visual SLAM, RealSense D455) generating real-time 3D elevation maps, cutting manual teleoperation time by 70%.
- Slashed CV pipeline latency by ~60% and raised throughput 4x via CUDA, YOLOv8, and ArUco/AprilTag detection across 4 simultaneous camera streams.
- Flashed Micro-ROS firmware to ESP32s; fused IMU and encoder data with an EKF, cutting odometry drift by ~40% and enabling sub-10 cm autonomous positioning.

Virtual Intern, Data Excellence Agent — Samsung PRISM May 2026 - Oct 2026
- Built within Phase 1 (Foundation) of an agentic, multi-agent data intelligence pipeline for dataset discovery, governance, and quality analysis, under the mentorship of Samsung R&D.

Project Intern, Living Solution Control Development — LG Soft India Pvt. Ltd. (HS/ES India Lab) 1 Jun 2026 - 26 Jun 2026
- Delivered a project assignment on living solution control development at LG Electronics' HS/ES India Lab; work recognized as technically competent by LGSI leadership.

Social Media Manager — Clinigo (VIT Incubated Startup) Sep 2023 - Dec 2023
- Planned and produced LinkedIn/Instagram content for a healthcare-digitization startup, driving a 70% increase in engagement.

Creya (STEAM) Teacher — GEMS International School Gurgaon Apr 2023 - Jul 2023
- Guided students (grades 2–8) through hands-on electronics and mechanical-engineering projects in the school's STEAM program.

PROJECTS

Gesture-Controlled 5-DOF Robotic Arm | C++, Arduino, MPU6050, Flex Sensors 2026
- Mapped real-time MPU6050 orientation and 2 flex-sensor readings to 5-DOF servo commands with <50 ms latency; trained a gesture classifier on 200+ samples, achieving 94% accuracy across 8 gestures.

HayaiOS — Bare-Metal RTOS | C, ARM Assembly, ARM Cortex-M4 2026
- Implemented a preemptive scheduler (1 ms tick), context switching in <50 assembly instructions, mutex/semaphore primitives, and a full HAL isolating register-level I/O.

sort-tui — Terminal Sorting Algorithm Visualizer (PyPI) | Python, curses 2026
- Developed a zero-dependency, curses-based visualizer covering 149 sorting algorithms with 7 rendering modes, audio pitch mapping, side-by-side comparison, and a headless benchmarking mode.

Captivity — Autonomous Captive-Portal WiFi Login Client (PyPI) | Python, Rust 2026
- Built a plugin-based daemon that detects, parses, and auto-logs into WiFi captive portals in <200 ms via HTTP-204 probing, D-Bus network monitoring, and an encrypted credential vault.

teleop-cursor — Cursor-to-ROS2 Teleoperation Node (PyPI) | Python, rclpy 2026
- Published a zero-hardware ROS2 teleop node mapping desktop cursor displacement to Twist velocity commands at 10 Hz, validated on TurtleBot3 in Gazebo.

Sokobot — Warehouse Robot Fleet Management Platform | FastAPI, React, PostgreSQL 2026
- Engineered a full-stack fleet-management platform with JWT auth/RBAC and a background simulation engine modeling robot lifecycle (idle → navigating → charging), including nearest-idle-robot task allocation with battery-reserve constraints.

3D Reconstruction — IMU-Enhanced KinectFusion | C++, OpenCV, Intel RealSense SDK 2025
- Stabilized KinectFusion TSDF reconstruction by fusing RealSense D455 depth and IMU data via a complementary filter, reducing tracking failures by ~50% across a 6-DOF trajectory.

PATENTS & PUBLICATIONS

Autonomous Radar-Guided Survivor Detection and Navigation System — Indian Patent Application No. 202641072249 (filed 2026).
- Sensor-fusion system (radar + ultrasonic + IMU) with probabilistic confidence-mapping for through-debris detection — no SLAM, camera, or LiDAR required.

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified AI Practitioner · AWS Certified Cloud Practitioner.
- Oracle Cloud Infrastructure 2025 Certified Data Science Professional · Certified Generative AI Professional.
- Amazon ML Summer School 2026 — Selected among 3,000 students from 1.3 lakh+ applicants across India.
- International Rover Challenge — 13th Position Globally (2025), 17th Position Globally (2026).
- TEDx Speaker, TEDxGEMSInternationalSchool — Gender Stereotyping: Does It Still Exist? (2022).

LEADERSHIP & ACTIVITIES

- Anchor & Design Domain Member, VIT Anchoring Club — anchored one of VIT's largest cultural-club collaborations.

- School President, GEMS International School (2022–23); Head of Media & Design, Concordia MUN (2022); President, Innovation Club (2021–22); School Football Team Captain.
- Fundraiser, WWF-India Young Conservationist Programme (Dec 2021 – Jan 2022).

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, SQL, Bash, Verilog, C#
Robotics & Autonomous Systems: ROS2, Nav2, RTAB-Map, SLAM, EKF/UKF, PID Control, mmWave Radar
Perception & ML: OpenCV, YOLOv8, RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, HuggingFace Transformers
Embedded, AI Agents & Infra: ARM Cortex-M, ESP32, FreeRTOS, Docker, Git, AWS, LangGraph